

FORGE

AI-Powered Study & Exam Readiness Platform

A personal student dashboard built for a B.Tech Computer Engineering student — combining AI tutoring, quiz generation, viva prep, and readiness analytics in one app.

Netlify Functions

Groq AI

Gemini Live

Supabase Postgres

Vanilla JS

1. Project Overview

Forge is a static-frontend, serverless-backend web app that acts as an all-in-one AI study companion for a B.Tech CE student. It tracks exam readiness with real metrics, generates AI-powered notes/quizzes/doubt-answers from Groq, offers a real-time voice viva-prep mode via Gemini Live, and includes a hardened admin/parent view backed by Supabase.

Two entry points ship in production: `index.html` (student dashboard) and `admin-dashboard.html` (admin/parent view), both served as static files from Netlify with no client-side framework or build step.

2. Core Features



Exam Readiness Engine

4-metric algorithm ("Am I Ready?") computing a real readiness score from actual study activity — not a static number.



Ask Forge (AI Doubt Chat)

Groq-powered doubt-solving chat, plus a dedicated course-mode chat tied to YouTube lecture context.



AI Notes Generator

Generates structured notes on demand and saves them to a personal "My Notes" library, with flashcard support.



AI Quiz Engine

Auto-generated quizzes with AI answer grading (`generate-quiz.js` , `grade-answer.js`).



Viva Prep (Beta)

Simulated viva Q&A grading, plus an opt-in **real-time voice mode** powered by the Gemini Live API (mic in → spoken AI response out).



Brainstorm Challenge (Beta)

Timed challenge mode with AI grading and a **real public leaderboard** aggregated live across every registered student account.



YouTube Course Mode

English-filtered lecture video finder per topic/chapter, feeding a course-specific chat context.



Analytics & Trends

Subject trend charts, topic pattern breakdown, 84-day activity heatmap, brainstorm pattern chart, exam history.



Courses & Timetable

Course list with chapter-level topic modals, official weekly class timetable, upcoming deadlines tracker.



Study Timer

Built-in focus/study timer feeding into activity tracking and the readiness engine.



Academic Overview

Attendance tracking, CGPA history, backlog subject tracking, weekly AI-suggested study strategy/plan.



Admin / Parent Dashboard

Real stat cards (registered/synced counts, course breakdown), account search, sync/restore/delete, gated by Netlify Edge Function Basic-Auth.

3. Tech Stack

Layer	Technology	Notes
Frontend	Static HTML / CSS / vanilla JavaScript	No framework, no build step — <code>index.html</code> + <code>admin-dashboard.html</code>
Backend	Netlify Functions (Node.js, CommonJS)	19 serverless functions in <code>netlify/functions/</code>
Edge Layer	Netlify Edge Functions (Deno)	<code>admin-gate.js</code> — Basic-Auth gate that works on the free plan (Pro-only <code>netlify.toml</code> auth fails silently)
AI — Text	Groq API	3-key failover chain (<code>_groq-client.js</code>) across <code>GROQ_API_KEY</code> / <code>_2</code> / <code>_3</code> on rate-limit
AI — Voice	Gemini Live API (<code>gemini-3.1-flash-live-preview</code>)	Real-time bidirectional voice for Viva Prep; ephemeral tokens minted server-side
Video Search	YouTube Data API	English-filtered lecture discovery
Database	Supabase (Postgres)	<code>accounts</code> table, RLS enabled, service-role/server-only access — migrated off Netlify Blobs (2026-08-08)
Rate Limiting	Netlify Blobs	30 requests/hour/IP, only remaining Blobs usage
Hosting	Netlify	Site <code>forge-do</code> , free/dev-team plan

4. Architecture Highlights

Serverless & Static-First

Zero build pipeline — HTML/JS served directly, functions deployed as individual Netlify Lambdas. Keeps deploys fast and infra minimal for a single-user-scale app.

AI Failover Design

Every AI-backed function routes through a shared Groq client that automatically retries across 3 API keys on rate-limit errors, verified across all 10 AI-facing functions rather than per-function ad hoc handling.

Security Hardening

- Admin routes gated by an Edge Function doing constant-time (SHA-256) Basic-Auth comparison, credentials in env vars — not hardcoded.
- Supabase RLS enabled with zero public policies (service-role-only, server-side access).
- Per-IP rate limiting (30 req/hr) on public-facing functions.

Real Data, Not Mocks

Readiness scores, leaderboard rankings, focus-line suggestions, and trend charts are all computed from genuine stored activity — a deliberate move away from earlier hardcoded/placeholder versions of the same UI.

5. Benefits

Benefit	Why it matters
All-in-one workflow	Doubts, notes, quizzes, viva prep, and readiness tracking live in one dashboard instead of scattered tools.
Real progress signal	The readiness engine and focus line are driven by actual study activity, giving an honest picture instead of a vanity metric.
Low operating cost	Serverless functions + free-tier hosting + Groq's fast inference keep this essentially free to run at personal scale.
Resilient AI layer	Multi-key Groq failover means a single rate-limited key doesn't take down doubt-solving, notes, or grading.
Voice-native viva practice	Gemini Live lets a student rehearse spoken answers, closer to a real viva than typed Q&A.
Social motivation	The real cross-student Brainstorm leaderboard adds light competitive pressure without exposing private data.
Parent visibility	A separate, properly access-gated admin view gives oversight without merging into the student's own UI.
No framework overhead	Plain HTML/JS keeps the codebase approachable and fast-loading, appropriate for the project's actual scale.

6. Known Tradeoffs & Accepted Risks

Deliberate, documented tradeoffs — not oversights

- Admin page's client-side login is UX only, not real security; the Edge Function Basic-Auth gate is the actual barrier.
- Admin API key and Basic-Auth password are necessarily present in client-side JS (same threat model as any static-site embedded secret) — accepted given personal-project scale.
- `sync-account` / `restore-account` have no admin-key gate — matches the app's deliberate "email-only, no password" account model.
- `docs/` folder is inside the published root; currently low-risk (specs/plans only) but flagged for a future redirect rule.

7. Roadmap / Not Yet Started

- B.Sc(H) AI & DS track — second onboarding program, blocked on a real subject/chapter list.
- Notes-in-chat — rendering Ask Forge answers as note cards directly in the chat UI.
- Groq token-remaining display — blocked on Groq not exposing a quota-remaining endpoint.
- Full end-to-end verification of the Viva Prep Live Voice Mode's mic-capture loop in a real browser (automation can't grant real mic permission).